Vibe Route

Google Maps tells you the fastest route. I built a tool that tells you the best one.

An AI Product Manager case study - from research to shipped product in 5 weeks.

4	6	30+	5
AI Components	Research Studies	POIs per Route	Weeks to Ship

vibe-route.vercel.app | React + Google Maps API + Claude

The Problem Nobody Has Solved

The backstory: A former Google Maps Senior UX Researcher explained why "scenic" routing is hard: scoring beauty can bias toward affluent neighborhoods.
Google distanced from this — but the feature still doesn't exist.

The Competitive Gap

Scenic Driving
✓ Solved

Roadtrippers
Scenic

Hiking & Trails
✓ Solved

AllTrails
Komoot

Urban Walking
✗ Huge gap

Google Maps
optimizes
only for speed

The feature is technically feasible. The real challenge is designing it without encoding socioeconomic bias.

Every Decision Backed by Research

6 peer-reviewed studies shaped the MVP.

1

Detour tolerance

Leisure walkers accept ~25% detours

→ **Cap detours at ~30%**

2

What people love

Parks, sky visibility, amenities

→ **Define vibe dimensions**

3

Facility detours

Walkers detour for attractive stops

→ **Waypoint injection**

4

Night safety gap

Different needs after dark

→ **"Night Walker" persona**

5

Avoid poor lighting

People avoid dark routes

→ **Well-Lit scoring (V2)**

6

Night attention shifts

Hazard scanning changes

→ **Night-mode weights**

So what? Research turned "vibe" into measurable signals and scoped what to ship first.

How Vibe Route Works

1. **Enter Origin & Dest** — Google Places Autocomplete
2. **Select Vibes** — Click chips or type natural language intent
3. **AI Discovers Waypoints** — ML clustering finds "vibe zones"
4. **Score Routes** — API queries parks/cafes along each path
5. **Generate Narratives** — LLM describes routes via verified POIs
6. **Compare & Navigate** — Review and open in Google Maps

NLP Input Example

User types: "quiet walk through parks with coffee stops"

AI detects:

🌳 Green & Peaceful

☕ Coffee Stops

What You Get

Google Maps today:

"3.9 km · 56 min"

Vibe Route:

"The Park Path — 4.1 km · 58 min — tree-lined walk past Trinity Bellwoods with 2 coffee shops"

AI Architecture

4 AI components - each one earns its place.

AI #1 Waypoint Discovery

ML clustering to create genuinely different route options.

AI #2 Natural Language

Parse "quiet walk with coffee" → structured vibe prefs.

AI #3 Route Narratives

LLM summaries constrained to verified POIs (anti-hallucination).

AI #4 Vibe Scoring

Density-based scoring with diminishing returns.

What I rejected (on purpose)

- ✕ Personalization (not enough user data yet)
- ✕ "Scenic" LLM scoring (high bias risk)
- ✕ AI soundtrack (feature creep)

Designing Around Bias

Bias isn't a reason not to build - it's a design constraint.

Goal: deliver "better walks" without encoding socioeconomic bias.

1 Objective signals

Use parks, POI density, road types — not subjective "beauty".

Why: reduces value-judgment bias.

2 User-defined vibe

"Green & peaceful" is a preference — not a neighborhood ranking.

Why: shifts agency to the user.

3 Audit for correlation

Planned: 50 routes across diverse Toronto neighborhoods. Correlate vibe scores with census data.

Action: publish results + adjust model if biased.

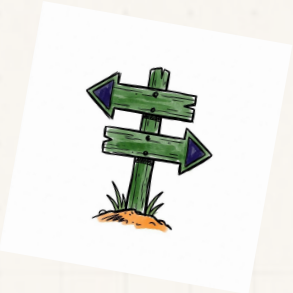
4 Reward local discovery

Count cultural spots + independent shops as positive "Local Character".

Why: rewards diversity, not just affluence.

Bottom line: I designed around the exact constraint that keeps mainstream maps from shipping "pleasantness".

Decisions & Tradeoffs



6 choices that shaped the MVP.

Overlap check

Alternates overlap → waypoint injection fallback.

Scoring curve

Linear → diminishing returns (log) to avoid score inflation.

Global-first MVP

No city datasets yet → ship everywhere using Maps APIs.

Progressive loading

Routes → scores → narratives for faster perceived speed.

Anti-hallucination

Constrained prompts + template fallback for reliability.

Scope cut

Defer "Well-Lit" to V2 due to data availability.

Tradeoff principle: ship the core loop first, then deepen quality + safety.

What I Shipped

A live product - not a concept deck.

MVP highlights

- ✓ Google Maps deep link (real routes)
- ✓ Vibe scoring + distinct alternatives
- ✓ LLM narratives from verified POIs
- ✓ Natural language input
- ✓ Day/night adjustments
- ✓ Mobile-responsive UI

Cost: $5–$22 total to build + launch (APIs + LLM during dev)

Try it: vibe-route.vercel.app

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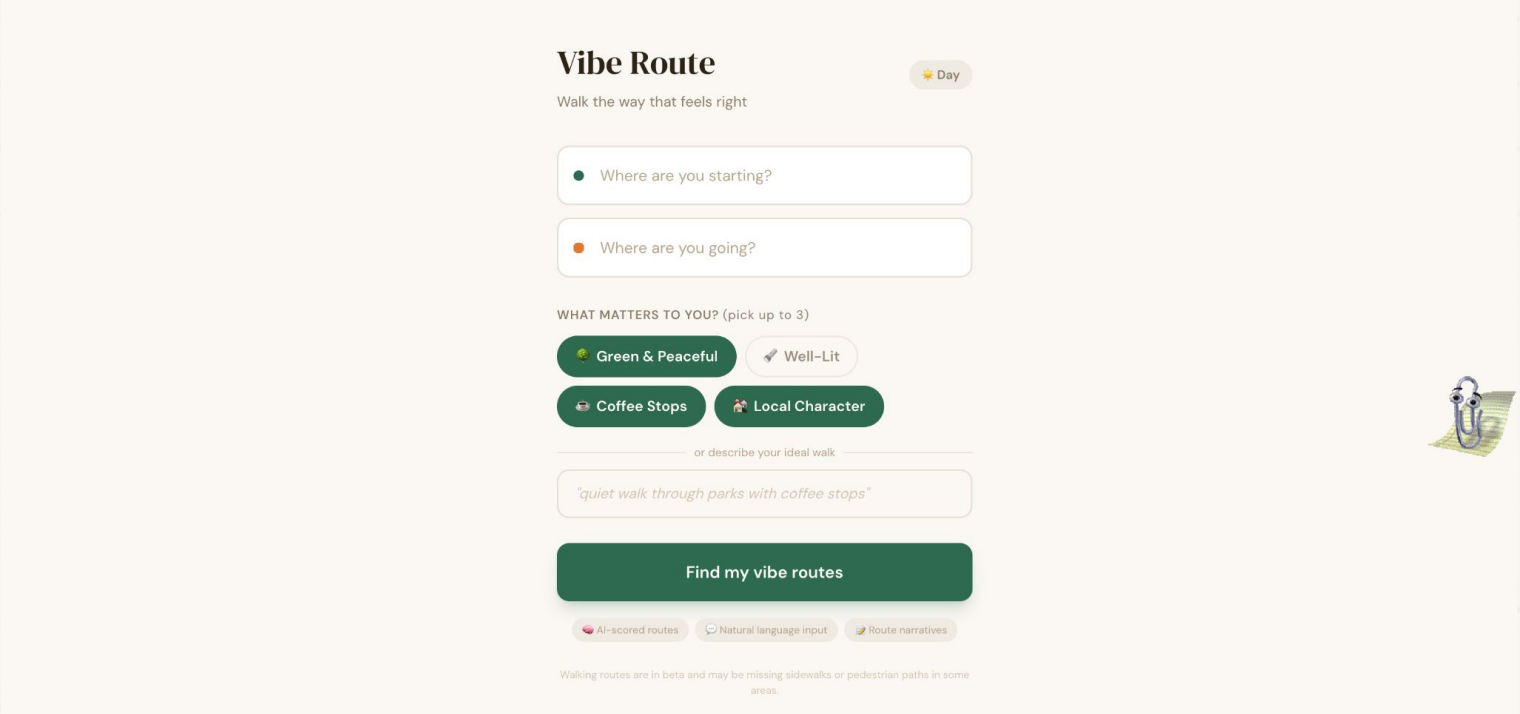

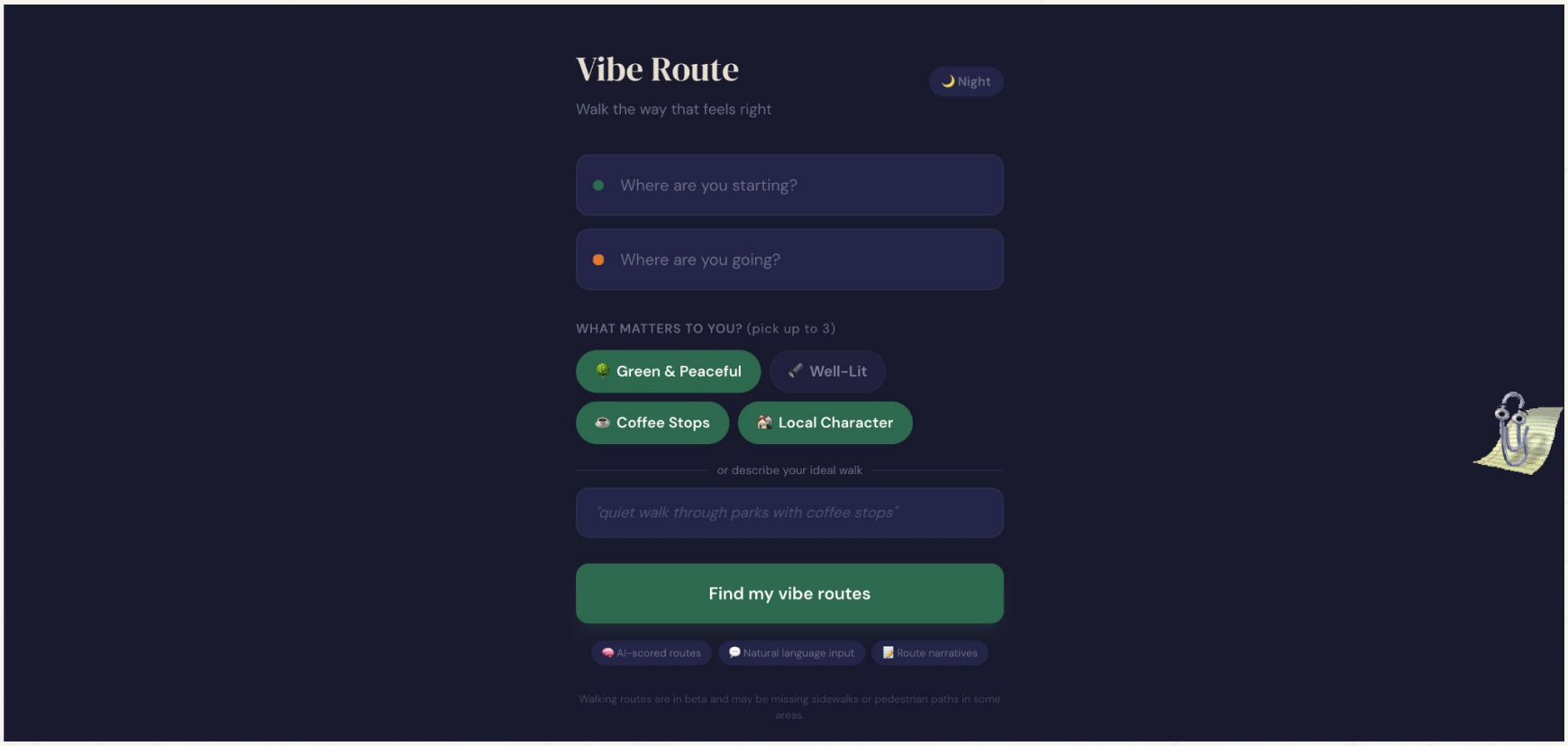

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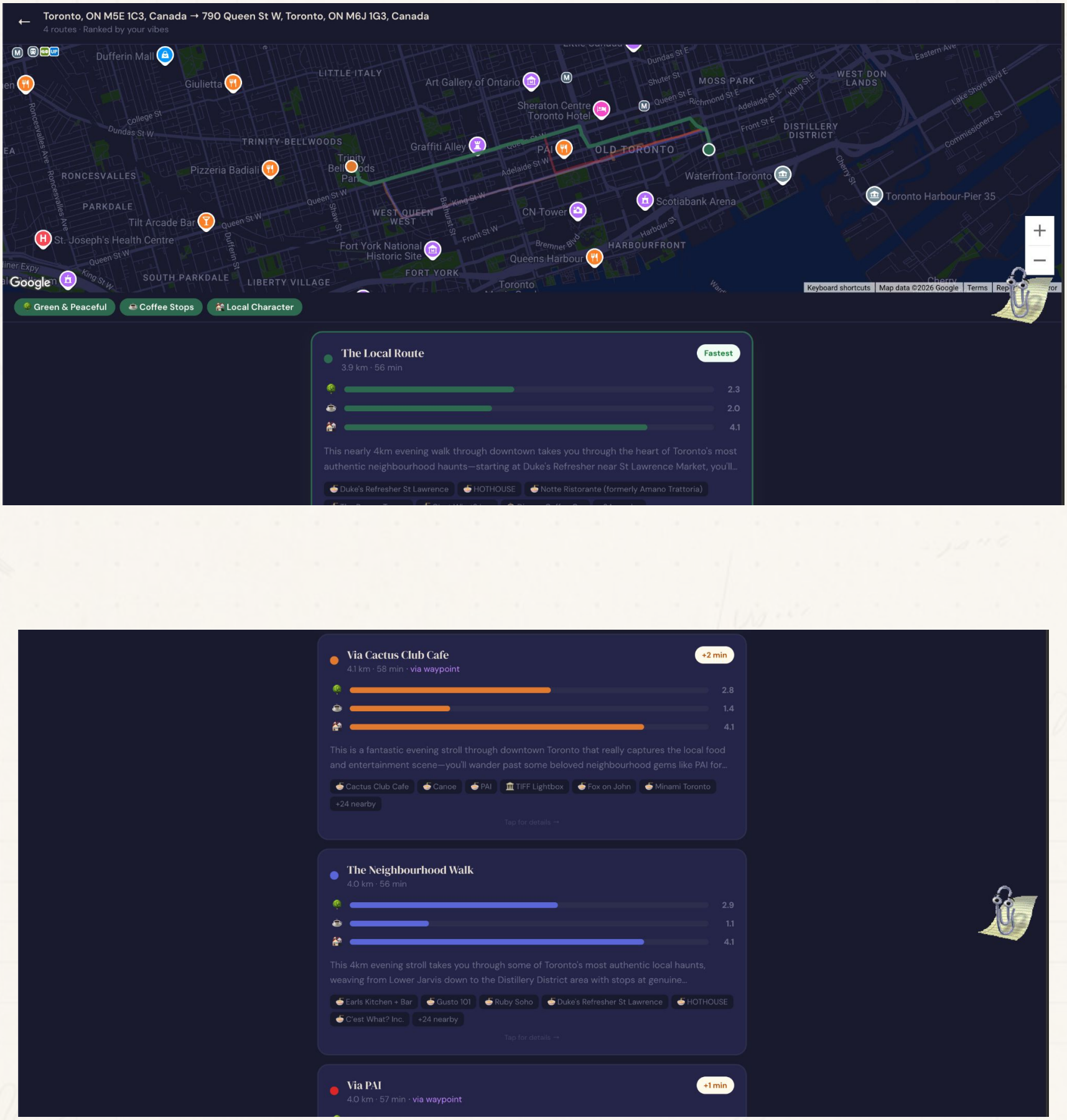

Success Metrics

What I planned to measure + what I already validated.

Pre-launch targets

Routes requested / session → > 1.5
Are users searching vs bouncing?

Vibe selection distribution
Green & Local > 40% each (persona validation)

Route card tap-through → > 30%
Are scores + narratives compelling?

Waypoint injection trigger → > 40%
Proves Google alts are too similar.

AI narrative load success → > 95%
(fallback)
Reliability in production.

Open in Google Maps → > 20%
Core conversion (choose + act).

Data I already have

Overlap is real
2/6 routes had 100% overlap; 2 had 40–50% overlap

Waypoint injection works
Different neighborhoods + distinct POIs (Toronto testing)

Log scoring separates routes
0.8–4.2 range (vs all hitting 5.0)

LLM accuracy audit
20 narratives checked; 0 hallucinated locations

Caching helps
~40% Places API cache hit rate

Post-launch tracking

- Vercel Analytics (privacy-friendly, no cookies)
- Custom events: searches, vibe selects, taps, Maps clicks
- No PII collected
- Weekly review (first month) → adjust weights/UI

Future Roadmap

What I'd build next (and why not yet).

V2 - Street View Computer Vision

Green View Index via semantic segmentation (measure what a route actually looks like).

V2 - Well-Lit Scoring

Use streetlight datasets to support the Night Walker persona.

V2 - Bias Audit

Run 50 routes across diverse neighborhoods, correlate with census income data, publish results.

V3 - Community Layer

User-submitted vibes + ratings. Big value — but comes with moderation & privacy risks.

Shipping > perfecting: MVP proves judgment - then we deepen safety + signal quality.

Key Takeaway

"The best AI project isn't the one with the most AI - it's the one where every AI component exists because the product is genuinely **better** with it (and you can explain why)."

What this project demonstrates

- ✓ Research-driven product thinking
- ✓ Honest about limitations
- ✓ AI applied where it adds value
- ✓ Bias-aware design
- ✓ End-to-end shipping
- ✓ Scope management

Try it: vibe-route.vercel.app

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